

Semantic Data Layer (SDL)

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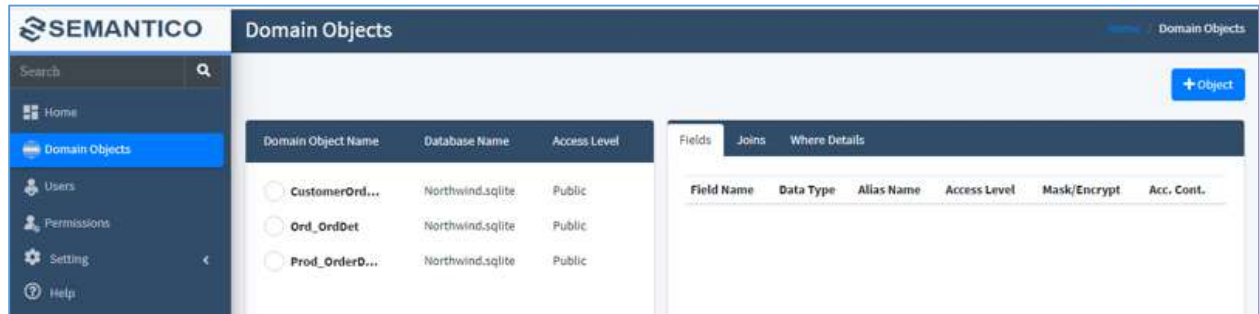
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Create Domain Object

To add a new domain object, follow the steps given below:

1. In the Semantico application, click **Domain Objects** > **+Object**.



2. The **Add Domain Object** screen appears as shown below.

The screenshot shows the 'Add Domain Object' screen in the Semantico application. The sidebar on the left has links for Dashboard, Domain Object (highlighted), User Management, and Permission. The main content area is titled 'Add Domain Object' and contains a 'Data Warehouse Connection Detail' section. This section includes a dropdown menu labeled 'Select', and three input fields: 'Server Name' (with placeholder 'Enter Server Name'), 'User Name' (with placeholder 'Enter User Name'), and 'Password' (with placeholder 'Enter Password'). At the bottom right of the form is a blue 'Next' button.

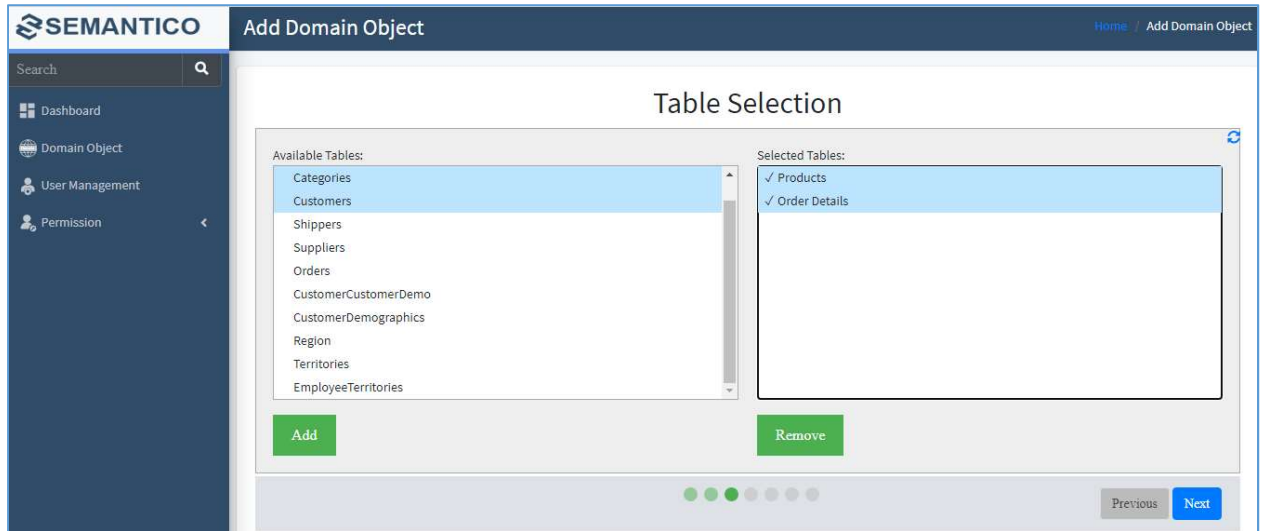
- a. Select one of the following servers.
 - SQL Server
 - DynamoDB
 - SQLITE

Note: Based on the server selection, the required fields are enabled.
 - b. Enter the Server Name.
 - c. Enter the User Name.
 - d. Enter the Password.
 - e. Click **Next**.
3. The **Domain Object Info** page appears.
 - a. Select the Database.
 - b. Click **Next**.

The screenshot shows the SEMANTICO application interface. The top navigation bar includes the SEMANTICO logo and the title 'Add Domain Object'. A sidebar on the left contains a search bar and navigation links for Dashboard, Domain Object, User Management, and Permission. The main content area is titled 'Domain Object Info' and features a 'Select Database' dropdown menu with a '--Select--' placeholder. At the bottom of the main area, there are five progress indicators (the first is green) and 'Previous' and 'Next' buttons.

4. The Table Selection page appears.
 - a. Select the available tables.
 - b. Click **Add** to add the tables to the Selected Tables.
 - c. Click **Remove** to remove the tables from the Selected Tables.
 - d. Click **Next**.

This screenshot shows the 'Table Selection' page within the SEMANTICO application. The layout is similar to the previous screenshot, but the main content area is titled 'Table Selection'. It contains two lists: 'Available Tables' and 'Selected Tables'. The 'Available Tables' list includes 'Employees', 'Categories', 'Customers', 'Shippers', 'Suppliers', 'Orders', 'Products', 'Order Details', 'CustomerCustomerDemo', and 'CustomerDemographics'. The first three items are checked. Below these lists are 'Add' and 'Remove' buttons. The progress indicators at the bottom show the first three steps as green, and the 'Next' button is visible.

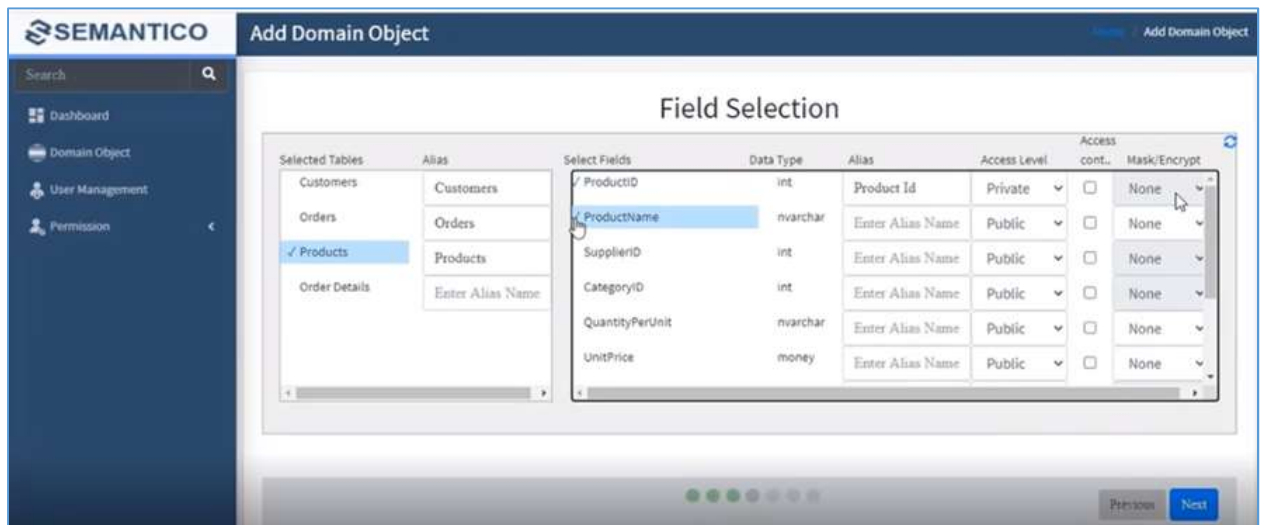


5. The **Field Selection** page appears.

a. Click the table on the left panel.

- Enter a unique alias name for the field which is common among the selected tables.
- Select the **Access Level**. Please refer the screenshot below for details.
 - **Public** - Name and Data Accessible to All.
 - **Private** - Name Viewable and Data Accessible for private users only.
 - **Protected** - Name Viewable and Data Accessible for protected users only.
- Select the **Access Control** check box.
 - For example, if you select Access Control for Customer Country. You are enabled to select the country in **Permission > Feature Level**.
 - Otherwise, all the countries are considered.
- Select the **Mask/Encrypt**.
 - **Mask** - to show asterisks.
 - **Encrypt** - to show encrypted script.

b. Click **Next**.



6. The **Table Join Details** page appears. This section is *optional*.
 - a. Select the *Left Table*.
 - b. Select the *Join*.
 - c. Select the *Right Table*.
 - d. Select the *Left Table Field*.
 - e. Select the *Right Table Field*.
 - f. Select the *Condition*.
 - g. Click **Add** to add the details.
 - h. Click **Next**.

SEMANTICO Add Domain Object

Search

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Table Join Details

Left Table: Order Details[OrderDetails] Join: Inner Join Right Table: Products[Products]

Left Table Field: --select-- Right Table Field: --select-- Condition: = **Add**

Left Table	Join Type	Right Table	Left Table Field	Right Table Field	Condition
Products	Inner Join	Order Details	ProductID	ProductID	=

Previous Next

7. The **Where Conditions** page appears.
 - a. Select the Table.
 - b. Select the Column.
 - c. Select the Condition.
 - d. Enter a Value.
 - e. Click **Add**.
 - f. Click **Next**.

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Where Conditions

Select Table: Products[] Select Column: --select-- Select Condition: --select--

Enter Value: Enter Value **Add**

Previous Next

8. The Domain Object Summary page is displayed.

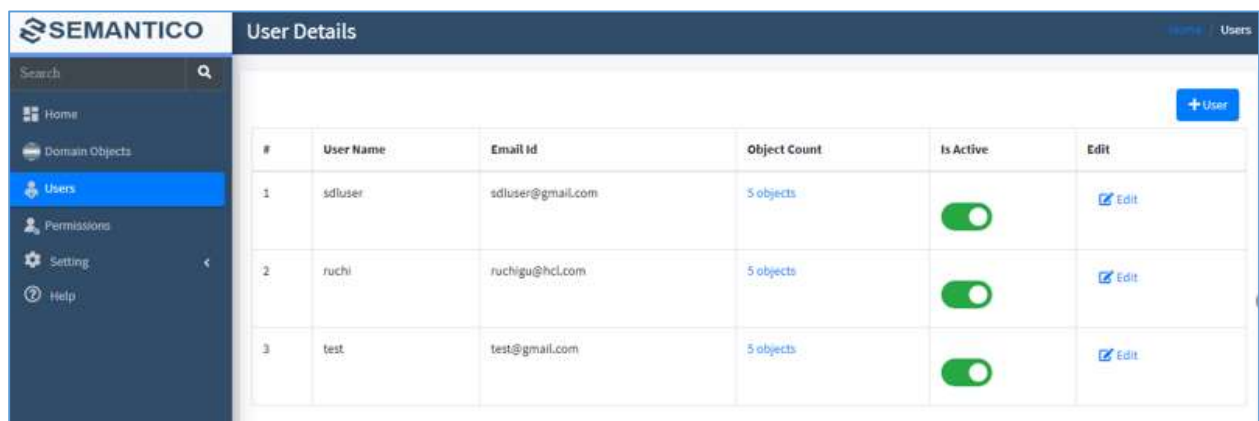
Semantic Data Layer

- a. Enter the Domain Object Name.
 - b. Select the Access Level.
 - c. Click **Submit**.
9. As a result, the domain object is added in the Domain Object List.

Users

The Users section is used to add new users and view the existing users.

1. Click **Users**.
2. You the view the list of users.
3. Click **+User** to add a new user.
 - a. Enter User Name.
 - b. Enter Email ID.
 - c. Enter Password.
 - d. Click **Submit**.
4. As a result, the user details will be added.



#	User Name	Email Id	Object Count	Is Active	Edit
1	sdluser	sdluser@gmail.com	5 objects	☒	Edit
2	ruchl	ruchigu@hcl.com	5 objects	☒	Edit
3	test	test@gmail.com	5 objects	☒	Edit



SEMANTICO Create User

User Name : Email Id :

Password : Key :

Grant Permission for Domain Object

Permission has to be granted to the user for accessing a domain object, its rows and columns. For details, refer the [Example based on Access Level](#).

To give permission to the domain object, follow the steps below.

1. Click **Permission > Object/Feature/Attribute Level**.
2. In Domain Object Level,
 - a. Select the **Username**.
 - b. Select the **Domain Object**.
 - c. Select the **User Access Type**.

- Select **Public** to access the domain and Field information that are set as Public.
 - In addition, protected fields will be displayed as 'None'.
 - Private Fields will be hidden.
- Select **Private** to access the domain and Field information that are set as Private.
 - All the fields will be shown irrespective of the conditions which also include Public and Protected.
- Select **Protected** to access the domain information that are set as Protected.
 - Protected Fields will be masked with asterisk symbols.
 - In addition, public fields will be shown.
 - Private Fields will be hidden.
- d. Select the **Permission**.
- e. Click **Execute** to give permission to the selected username.
- f. Click **Next**.

SEMANTICO Grant Row-Column Level Permissions

Home / Object/Feature/Attribute Level Permissions

Domain Object level Feature level Attribute level

UserName : sdluser Domain Object : Device Data

User Access Type : Private Permission : Allow

Execute Next

3. In the Feature Level tab, follow the steps below to provide row level permission.
 - a. Select the **Username**.
 - b. Select the **Domain Object**.
 - c. Select the **Fields**.
 - d. Select the **Field Values**.
 - e. Click **Execute**.
 - f. Click **Next**.

SEMANTICO Grant Row-Column Level Permissions

Home / Object/Feature/Attribute Level Permissions

Domain Object level Feature level Attribute level

UserName : sdluser Domain Object : Prod_OrderDetail Fields : Quantity

Field Values:

- ✓ 1
- 10
- 100
- 11
- 110
- 12
- 120
- 13
- 130
- 14

Execute Next

4. In the Attribute Level, follow the steps below to provide column level permission.

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- Select the **Username**.
- Select the **Domain Object**.
- Select the **Fields**.
- Select **Execute**.

SEMANTICO Grant Row-Column Level Permissions [Home](#) / [Object/Feature/Attribute Level Permissions](#)

Search

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Object/Feature/Attribute Level

Domain Object level Feature level Attribute Level

UserName : sdluser **Domain Object :** Device Data **Access Level :** Public

Fields:

✓ headername	Public
✓ headertype	Public
✓ header-deviceid	Public
✓ Id	Public
✓ msg_activeThreads	Public
✓ deviceTypeName	Public
✓ deviceid	Public
✓ deviceSerialNo	Public

Execute

Example based on Access Level

Based on the access level, mask/encryption and access control, the fields are shown to the users as per the access they have.

The below table has a combination of Access Levels and the other tables that follow show the end result based on the access the user has.

Field Name	Data Type	Alias Name	Access Level	Mask/Encrypt	Acc. Cont.
header.name	String	header.name	Public	None	FALSE
header.type	String	header.type	Public	Encrypt	FALSE
header.device-id	String	header.device-id	Protected	None	FALSE
Id	String	Id	Private	None	FALSE
message.activeThreads	Numeric	message.activeThreads	Public	None	FALSE
message.getProcessCpuLoad	Numeric	message.getProcessCpuLoad	Public	None	FALSE
device.deviceType.active	BOOL	device.deviceType.active	Public	None	FALSE
device.deviceType.name	String	device.deviceType.name	Public	None	TRUE
device.name	String	device.name	Public	Mask	FALSE
device.status	BOOL	device.status	Public	None	FALSE

Access Level - Public

User-sdluser			Access Level - Public					
header.name	header.type	header.device-id	message.activeThreads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	Android	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE

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monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE

Result: Private fields are not visible; the fields are masked and encrypted as selected from the dropdown.
Protected field is showing NULL value.

Access Level - Protected

User-sdluser			Access Level- Protected					
header.name	header.type	header.device-id	message.activeThreads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	*****	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	*****	53	0	FALSE	Android	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	*****	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	*****	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	*****	53	0	FALSE	IOS	*****	FALSE

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monitor 2	bqLRic3y6rC57ulbHa+icg==	*****	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	*****	53	0	FALSE	Android	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	*****	53	0	FALSE	IOS	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	*****	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	*****	53	0	FALSE	Android	*****	TRUE

Result- Private fields are not visible; the fields are masked and encrypted as selected from the dropdown.
Protected field is masked.

Access Level - Private

User-sdluser			Access Level- Private						
header.name	header.type	header.device-id	Id	message.active Threads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	9	53	0	FALSE	Android	LG Phones	TRUE
monitor2	DATA2	968d731b-4511-47b1-846c-bb54458d7899	1	53	0	FALSE	Android	AD681H	FALSE
monitor2	DATA2	968d731b-4511-	4	53	0	FALSE	IOS	AirPods Pro	TRUE

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		47b1-846c-bb54458d7899							
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	3	53	0	FALSE	Android	Lenovo Tab M10	TRUE
monitor2	DATA2	968d731b-4511-47b1-846c-bb54458d7899	5	53	0	FALSE	IOS	Apple Watch Series 4	FALSE
monitor2	DATA2	968d731b-4511-47b1-846c-bb54458d7899	7	53	0	FALSE	IOS	iPad	TRUE
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	1 0	53	0	FALSE	Android	AMTC	TRUE
monitor1	DATA1	968d731b-4511-47b1-846c-	2	53	0	FALSE	IOS	AirPods	TRUE

Semantic Data Layer

		bb54458d7899							
monitor2	DATA2	968d731b-4511-47b1-846c-bb54458d7899	6	53	0	FALSE	IOS	Apple TV	TRUE
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	8	53	0	FALSE	Android	Samsung M30	TRUE

Results - All private fields are visible, and fields are unmasked and unencrypted.

Feature Level Permission

User : sdluser	device.deviceType.name : Android	User access level : Private
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Grant Row-Column Level Permissions

Object/Feature/Attribute Level Permissions

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User Management

Permission

Object/Feature/Attribute Level

Domain Object level

Feature level

Attribute Level

UserName :

sdluser

Domain Object :

Sample Object

Fields :

device.deviceType.name

Field Values:

☒ Android

☐ IOS

Execute

Next

header.name	header.type	header.device-id	Id	message.active Threads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	9	53	0	FALSE	Android	LG Phones	TRUE

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monitor2	DATA2	968d731b-4511-47b1-846c-bb54458d7899	1	53	0	FALSE	Android	AD681H	FALSE
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	3	53	0	FALSE	Android	Lenovo Tab M10	TRUE
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	1 0	53	0	FALSE	Android	AMTC	TRUE
monitor1	DATA1	968d731b-4511-47b1-846c-bb54458d7899	8	53	0	FALSE	Android	Samsung M30	TRUE

Result - Record having "Android" field will be only shown.

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User - rajesh			User access level - Public					
header.name	header.type	header.device-id	message.activeThreads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	Android	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	53	0	FALSE	Android	*****	TRUE

Attribute Level Permission

User : rajesh	User access level : Public	Field selected as Private : Id
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SEMANTICO Grant Row-Column Level Permissions

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Object/Feature/Attribute Level

Domain Object level Feature level Attribute Level

Username : pratham Domain Object : Sample Object Access Level : Public

Fields:

- ✓ header.name Public
- ✓ header.type Public
- header.device-id Protected
- ✓ Id Private
- ✓ message.activeThreads Public
- ✓ message.getProcessCpuLoad Public
- ✓ device.deviceType.active Public
- ✓ device.deviceType.name Public
- ✓ device.name Public

Execute

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header.name	header.type	header.device-id	Id	message.activeThreads	message.getProcessCpuLoad	device.deviceType.active	device.deviceType.name	device.name	device.status
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	9	53	0	FALSE	Android	*****	TRUE

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monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	1	53	0	FALSE	Android	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	4	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	3	53	0	FALSE	Android	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	5	53	0	FALSE	IOS	*****	FALSE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	7	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	10	53	0	FALSE	Android	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	2	53	0	FALSE	IOS	*****	TRUE
monitor 2	bqLRic3y6rC57ulbHa+icg==	NULL	6	53	0	FALSE	IOS	*****	TRUE
monitor 1	3aHxxhbQNxMe8TI9HJ32QA==	NULL	8	53	0	FALSE	Android	*****	TRUE

Result- If you are provided access for private field, then Private field will be visible.

In Attribute level permission, public fields are disabled for selection by default. Only Private or Protected fields are enabled. Although Protected fields are visible in all the three user access levels whether selected or not. But we can allow or disallow Private field from here for visibility.

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Grant Row-Column Level Permissions

Object/Feature/Attribute Level Permissions

Search

Dashboard

Domain Object

User Management

Permission

Object/Feature/Attribute Level

Domain Object level

Feature level

Attribute Level

UserName :

sduser

Domain Object :

Sample Object

Access Level :

Private

Fields:

✓ header.name	Public
✓ header.type	Public
header.device-id	Protected
id	Private
✓ message.activeThreads	Public
✓ message.getProcessCpuLoad	Public
✓ device.deviceType.active	Public
✓ device.deviceType.name	Public
✓ device.name	Public

Execute

Semantic Data Layer

Exceptions in Field Selection

Data Type - For Numeric and Boolean type fields, Mask/Encrypt dropdown is disabled.

SEMANTICO Add Domain Object [Home](#) / [Add Domain Object](#)

Search

- Dashboard
- Domain Object
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Field Selection

Selected Tables	Alias	Select Fields	Data Type	Alias	Access Level	Access cont..	Mask/Encrypt
✓ Sample	Enter Alias Name	header.device-id	S	Enter Alias Name	Public	☐	None
		Id	S	Enter Alias Name	Public	☐	None
		message.activeThreads	N	Enter Alias Name	Public	☐	None
		message.getProcessCpuLoad	N	Enter Alias Name	Public	☐	None
		message.usedMemory	N	Enter Alias Name	Public	☐	None
		message.getTotalPhysicalMe	N	Enter Alias Name	Public	☐	None
		memorySize	N	Enter Alias Name	Public	☐	None

Previous Next